

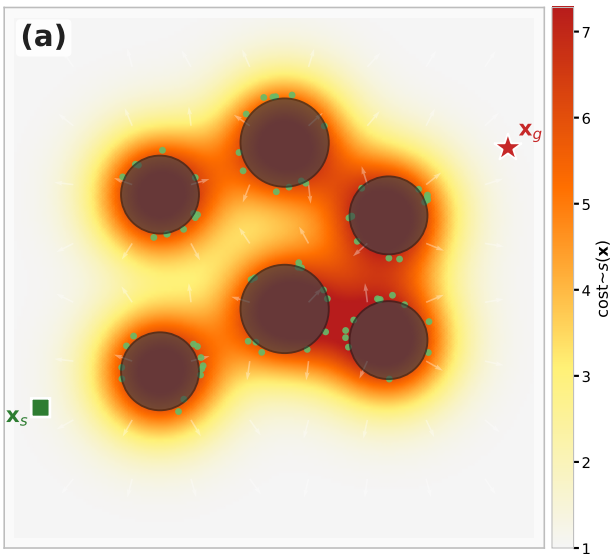
RIT*: Riemannian Informed Trees for Anisotropic Motion Planning

Riemannian Informed Set $\mathcal{I}_R \subset \mathcal{I}_E$

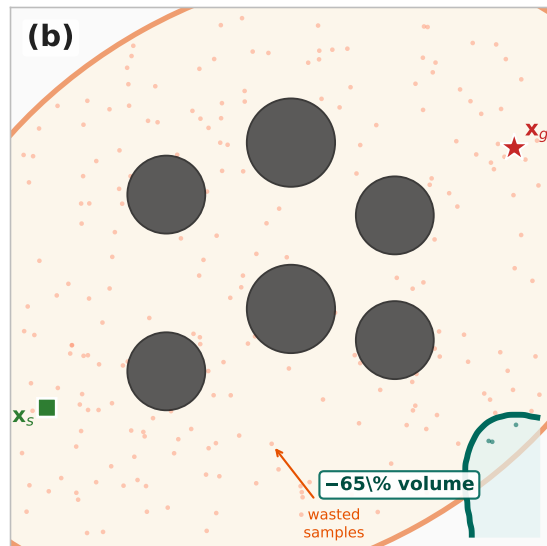
Path Quality: Cost-Aware vs. Euclidean

6-DOF $\mathcal{C}$ -Space: Informed Set Shrinkage

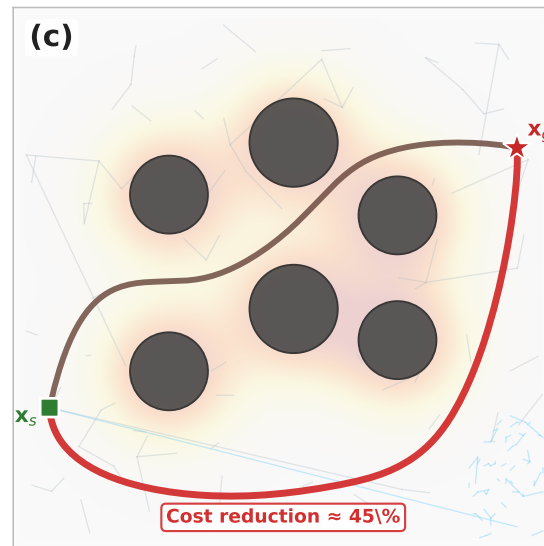
CARM: Collision-Adaptive Riemannian Metric



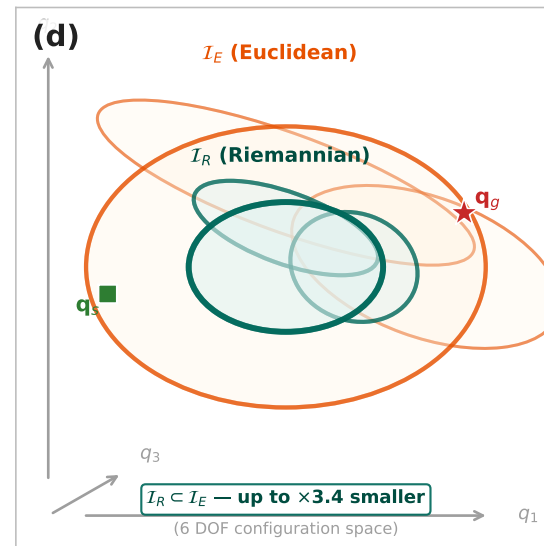
Online learning from collision feedback



Tighter search region $\rightarrow$ fewer samples needed



CARM routes around high-cost obstacle zones



$\kappa = 12$ inertia ratio $\Rightarrow 3.4 \times$ volume reduction

▭ Euclidean $\mathcal{I}_E$ (BIT*)
 ▭ Riemannian $\mathcal{I}_R$ (RIT*)
 — Euclidean path
 — RIT* path (CARM)
 — Euclidean tree
 — RIT* tree
 ● Collision feedback (CARM)